Chapter 3: General Measurements

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1. Basics of general measurements

Description of measurement:
Measurements represent the interface between quantum and classical information

- Performing a measurement on a system extracts classical information from the quantum state
- In general, the system is changed in the process.

First, we focus on destructive measurement, which produces a classical output.

2 ways of describing destructive measurement

1. As a collection of matrices, one for each measurement
2. A channel whose outputs are always classical states (represented by a diagonal density matrix).

Measurement as matrix:
Suppose we have a system X to measure. The set of classical states of X is $\{0, 1, \ldots, n-1\}$. The set of outputs is $\{0, 1, \ldots, m-1\}$

Projective measurement:
A projection measure is described as a collection of projection matrices $\{\Pi_0, \Pi_0, \ldots, \Pi_{n-1}\}$ that satisfy the following condition:

$$\Pi_0 + \Pi_1 + \ldots + \Pi_{n-1} = \mathbb{I}_X$$

IF the state of X is $|\psi\rangle$, each output a appears with probability :

$$Tr(\Pi_a |\psi\rangle\langle\psi|)$$

If the output state of X is ρ (density matrix), each output a appears with probability :

$$Tr(\Pi_a \rho)$$

A general measure is described as a collection of positive semi-definite matrices $\{P_0, P_0, \ldots, P_{n-1}\}$ that satisfy the following condition:

$$P_0 + P_1 + \ldots + P_{n-1} = \mathbb{I}_X$$

If the output state is ρ, each output appears with probability a:

$$Tr(P_a \rho)$$

This necessarily results in a probability vector $(Tr(P_0\rho), \ldots, Tr(P_{m-1}\rho))$. These are positive real numbers $Q, R \geq 0 \rightarrow Tr(QR) \geq 0$. The sum of these numbers is 1:

$$Tr(P_0\rho) + \ldots + Tr(P_{m-1}\rho) = Tr(\rho) = 1$$

<u>Example 1</u>: (any projective measurement)

Measure in the standard base can be represented as follows:

$$P_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \qquad P_1 = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \, 0 & 1 \end{pmatrix}$$

Measuring a qubit in the ρ state gives an output with probability :

$$P(output = 0) = Tr(P_0\rho) = \langle 0|\rho|0\rangle$$

$$P(output = 1) = Tr(P_1\rho) = \langle 1|\rho|1\rangle$$

<u>Example 2</u>: (a non-projective qubit measurement)

We pose P_0, P_1 :

$$P_0 = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

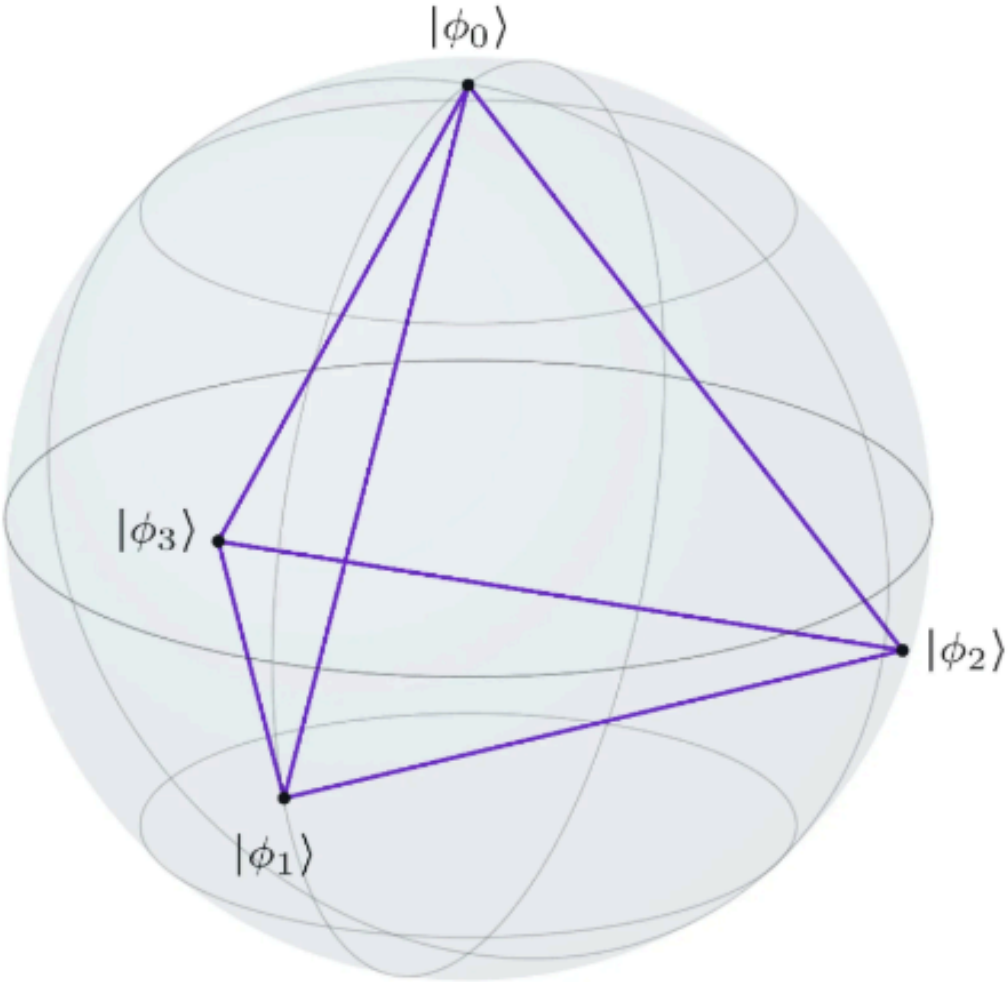
$$P_1 = \begin{pmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

Measuring a qubit in the $|+\rangle$ state gives the following output probabilities:

$$P(outcome = 0) = Tr(P_0|+\rangle\langle +|) = Tr\left(\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}\right) = \frac{5}{6}$$

$$P(outcome = 1) = Tr(P_1|+\rangle\langle +|) = Tr\left(\begin{pmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}\right) = \frac{1}{6}$$

<u>Example 3</u>: tetrahedral measure



Represents a tetrahedron in Bloch's sphere. The measures are defined as follows:

$$P_0 = \frac{|\phi_0\rangle\langle\phi_0|}{2}, P_1 = \frac{|\phi_1\rangle\langle\phi_1|}{2}, P_2 = \frac{|\phi_2\rangle\langle\phi_2|}{2}, P_3 = \frac{|\phi_3\rangle\langle\phi_3|}{2}$$

Mesaurements as channels

- Classical states are represented by a diagonal density matrix.

Any measurement can be described as a Φ channel:

- System input X is the system to be measured
- The classical output states Y are the possible measurements $\{0, 1, \ldots, m-1\}$.
- For any input ρ of X, the output state $\Phi(\rho)$ is a diagonal density matrix.

Example in the standard measurement basis

The complete phase-shift channel Δ is described in the standard qubit basis:

$$\Delta(\rho) = \langle 0|\rho|0\rangle|0\rangle\langle 0| + \langle 1|\rho|1\rangle|1\rangle\langle 1|$$

represents a form of decoherence.

Equivalence with a matrix description:

A Φ channel from X into Y has the property that $\Phi(\rho)$ is always diagonal if:

$$\Phi(\rho) = \sum_{a=0}^{m-1} Tr(P_a \rho)|a\rangle\langle a|$$

for measurements $\{P_0, P_0, \ldots, P_{m-1}\}$.

Partial measurements:

Suppose we have a system (X, Z) in state ρ and a measurement $\{P_0, P_0, \ldots, P_{n-1}\}$ is performed on X. The result is an output measure and the state of Z can change depending on the output.

Output probabilities:
The probabilities for different probabilities of output measures depend only on the measure and the reduced ρ_X state of X.

$$P(outcome = a) = Tr(P_a \rho_X) = Tr(P_a Tr_Z(\rho)) = Tr((P_a \otimes \mathbb{I}_Z)\rho)$$

States conditioned by output measurements: To determine the state of Z conditioned on a given measure, we can return to the channel description on the measure:

$$\Phi(\sigma) = \sum_{a=0}^{m-1} Tr(P_a \sigma)|a\rangle\langle a|$$

Applying this channel to X, we obtain the following result:

$$\sum_{a=0}^{m-1} |a\rangle\langle a| \otimes Tr_X((P_a \otimes \mathbb{I}_Z)\rho)$$

The state of Z conditioned by the output a can be obtained by normalizing the matrix $Tr_X((P_a \otimes \mathbb{I})\rho)$.

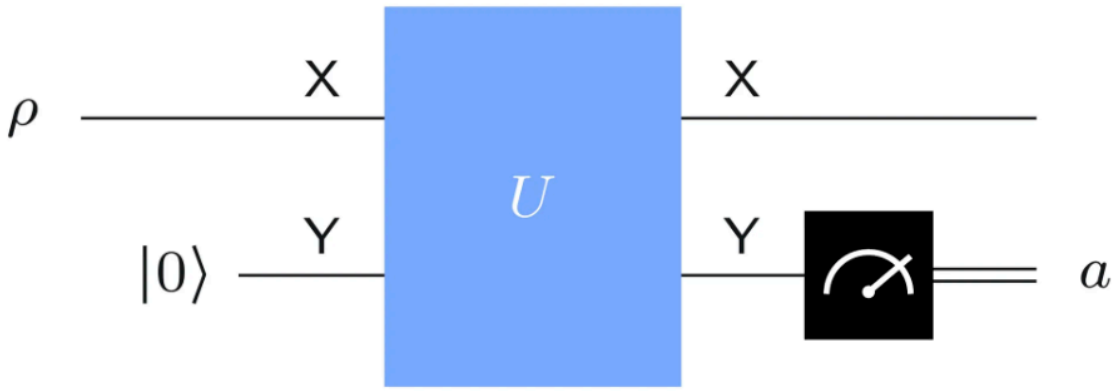
In short, when a $\{P_0, P_0, \ldots, P_{n-1}\}$ measurement is performed on X when (X, Z) is in a ρ state this occurs:

1. Each output a appears with probability $Tr_X((P_a \otimes \mathbb{I})\rho)$.
2. Conditional on obtaining the output a, the state of Z becomes:

$$\frac{Tr_X((P_a \otimes \mathbb{I})\rho)}{Tr((P_a \otimes \mathbb{I})\rho)}$$

2. Naimark's Theorem

Statement and proof:
Naimark's theorem is a fundamental theorem concerning measurements. It establishes that any general measure can be implemented in the following way:



Given a general measure $\{P_0, P_0, \ldots, P_{n-1}\}$ on X, the latter can be implemented as follows:

1. Introduce an initial workspace Y with classical states $\{0, 1, \ldots m-1\}$.
2. Perform a unit operation U on the pair (Y, X).
3. Perform a standard basis measurement on Y.

For each positive semi-definite matrix P, there is a unique semi-definite matrix Q such that $Q^2 = P$. This matrix is denoted $\sqrt{P}$. We can calculate $\sqrt{P}$ with the spectral decomposition of P:

$$P = \sum_{k=0}^{n-1} \lambda_k |\psi_k\rangle\langle\psi_k| \rightarrow \sqrt{P} = \sum_{k=0}^{n-1} \sqrt{\lambda_k} |\psi_k\rangle\langle\psi_k|$$

Any unit matrix U that follows has the following pattern:

$$U = \begin{pmatrix} \sqrt{P_0} & \ldots \\ \sqrt{P_1} & \ldots \\ \vdots & \vdots \\ \sqrt{P_{m-1}} & \ldots \end{pmatrix}$$

We need to check two things (1) such a matrix U works correctly, (2) such a matrix is unitary:

$$\sigma = U(|0\rangle\langle 0| \otimes \rho)U^\dagger = \begin{pmatrix} \sqrt{P_0}\rho\sqrt{P_0} & \ldots & \sqrt{P_0}\rho\sqrt{P_{m-1}} \\ \vdots & \vdots & \vdots \\ \sqrt{P_{m-1}}\rho\sqrt{P_0} & \ldots & \sqrt{P_{m-1}}\rho\sqrt{P_{m-1}} \end{pmatrix}$$

$$= \sum_{a,b=0}^{m-1} |a\rangle\langle b| \otimes \sqrt{P_a}\rho\sqrt{P_b}$$

$$\sigma_Y = \sum_{a,b=0}^{m-1} Tr_Y(\sqrt{P_a}\rho\sqrt{P_b})|a\rangle\langle b|$$

$$Prob(outcome = a) = \langle a|\sigma_Y|a\rangle = Tr(\sqrt{P_a}\rho\sqrt{P_a}) = Tr(P_a\rho)$$

Fonciton 1 ok

To check that U is unitary, we need to verify that each column vector of u forms an orthonormal basis. These column vectors are $|\gamma_0\rangle, \ldots, |\gamma_{n-1}\rangle$:

$$|\gamma_c\rangle = \sum_{a=0}^{m-1} |a\rangle \otimes \sqrt{P_a}|c\rangle \quad \langle\gamma_c|\gamma_d\rangle = \langle c|(\sum_{a=0}^{m-1} |\sqrt{P_a}\sqrt{P_a})|d\rangle = \langle c|d\rangle$$

function 2 ok.

Non-destructive measurement:
A non-destructive measurement not only has a classical output measurement, but also a post-measurement output state on the system being measured.

Non destructive measurement from Naimark's theorem:
Consider a general destructive measurement $\{P_0, P_0, \ldots, P_{n-1}\}$ on a system X. We can define a non-destructive measure with the same output probability using Naimark's theorem.

Conditioned on the output a of state X becomes:

$$\frac{\sqrt{P_a}\rho\sqrt{P_a}}{Tr(P_a\rho)}$$

Non destructive measuremetn from Kraus matrices:
Suppose $M_0, \ldots, M_{m-1}$ are square matrices satisfying the equation :

$$\sum_{a=0}^{m-1} = M_a^\dagger M_a = \mathbb{I}$$

They define a non-destructive measurement. For a system in the ρ state:

$$Pr(outcome = a) = Tr(M_a\rho M_a^\dagger) = Tr(M_a M_a^\dagger \rho)$$

Conditioned on the output den a, the measurement state of the output becomes:

$$\frac{M_a\rho M_a^\dagger}{Tr(M_a\rho M_a^\dagger)}$$

POVM

3.State discrimination & tomography

Quantum state discimination:
Let $\rho_0, \ldots, \rho_{m-1}$ be states of the system X and let $(p_0, \ldots, p_{m-1})$ be a probability vector.

- An element $a \in \{0, \ldots, m-1\}$ is chosen randomly with probabilities

$(p_0, \ldots, p_{m-1})$.

- The system X is prepared in the state ρ_a.
- Goal: determine the measure of X

Quantum Tomography Assume ρ to be an unknown quantum state of the system:

- Identical systems $X_1, X_2, \ldots, X_N$ are each prepared independently in the ρ state.
- Goal: approximate ρ by measuring $X_1, X_2, \ldots, X_N$.

State pair discrimination:
When $m = 2$ for state discrimination, the goal is to differentiate a pair of states. A pair of states is discriminated by the Helstron measure.

This is a projection measure on $\{\Pi_0, \Pi_1\}$ defined as follows.